

Answer: C

- 7 Work done by frictional force = change in gravitational potential energy + change in kinetic energy

$$\begin{aligned} W &= (GPE_f - GPE_i) + (KE_f - KE_i) \\ &= -(1.5 \times 9.81 \times 0.85 \sin 10^\circ) + (0 - \frac{1}{2} \times 1.5 \times 5.0^2) \\ &= -21 \text{ J} \end{aligned}$$

Answer: B